INSTRUCTOR SOLUTION

ID: ____________________

Habib University

EE-172/CS-130/CE-222 Digital Logic and Design – Spring 2026

Time = 30 minutes **Quiz 03** **Max Points: 20**

This Quiz assesses CLO No 2 – *"Apply principles of Boolean Algebra to represent and build equivalent realizations of digital logic (circuits)"*

Question 1 [4 + 4 points]:

Implement the following Boolean function F, together with don't-care conditions d by finding the following for gate-level implementation: -

(a) Simplified SOP expression.

(b) Simplified POS expression.

$$F(w,x,y,z) = \sum(0,2,3,8,10,11,13,14) \text{ and } d(w,x,y,z) = \sum(12,15)$$

Note – use the 4x4 map below by marking all necessary information

$wx \backslash yz$	00	01	11	10
00	m_0: 1	m_1: 0	m_3: 1	m_2: 1
01	m_4: 0	m_5: 0	m_7: 0	m_6: 0
11	m_{12}: X	m_{13}: 1	m_{15}: X	m_{14}: 1
10	m_8: 1	m_9: 0	m_{11}: 1	m_{10}: 1

wx

$x'z'$

$x'y$

a) $F = wx + x'y + x'z'$

SOP

b) $F' = w'x + x'y'z$

$\Rightarrow F = (w'x + x'y'z)'$

$= (w'x)' \cdot (x'y'z)'$

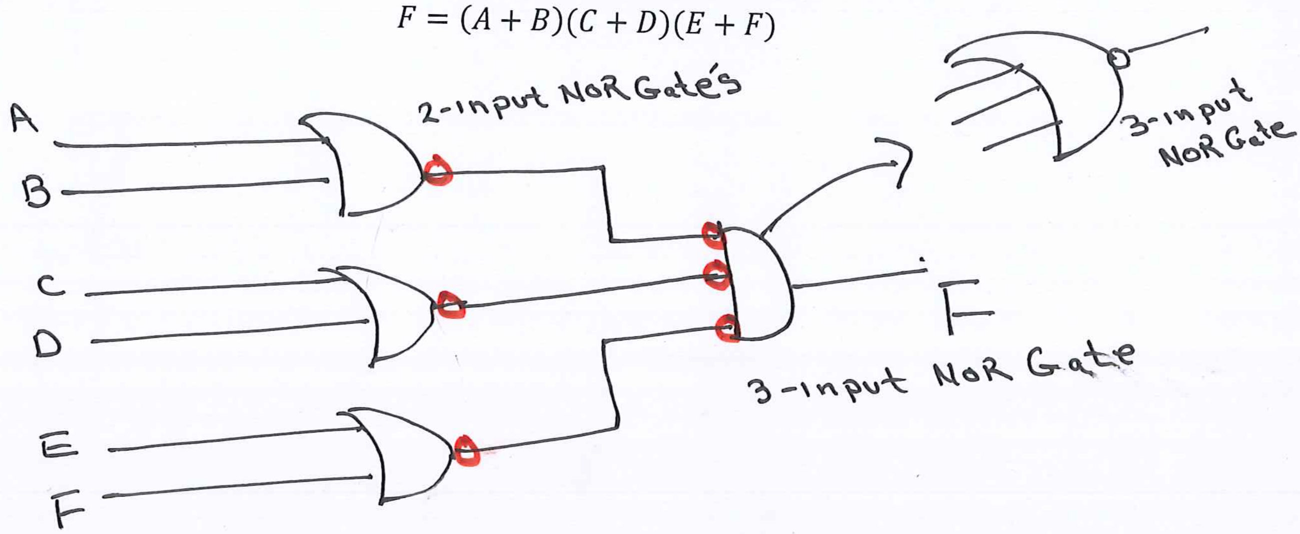
$F = (w+x') \cdot (x+y+z')$

POS

Question 2 [4 + 4 points]:

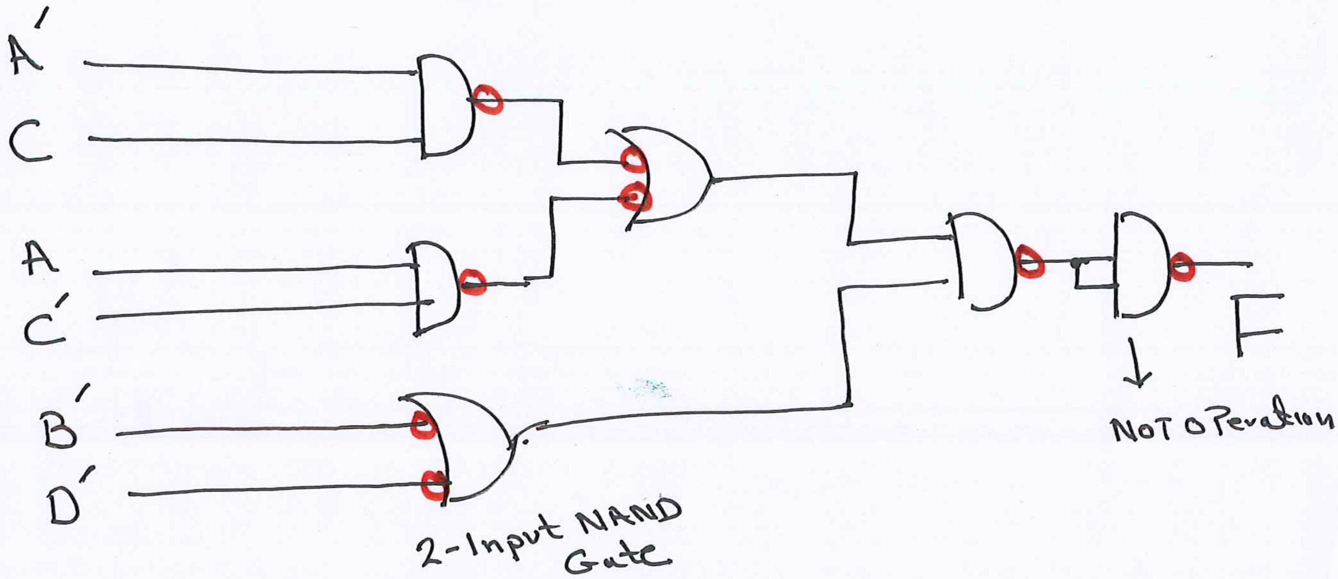
(a) Implement the following Boolean expression (expressed as POS) using NOR gates (only):

$$F = (A + B)(C + D)(E + F)$$



(b) Implement the following multi-level Boolean function using NAND gates (only):

$$F = (A'C + AC')(B + D)$$



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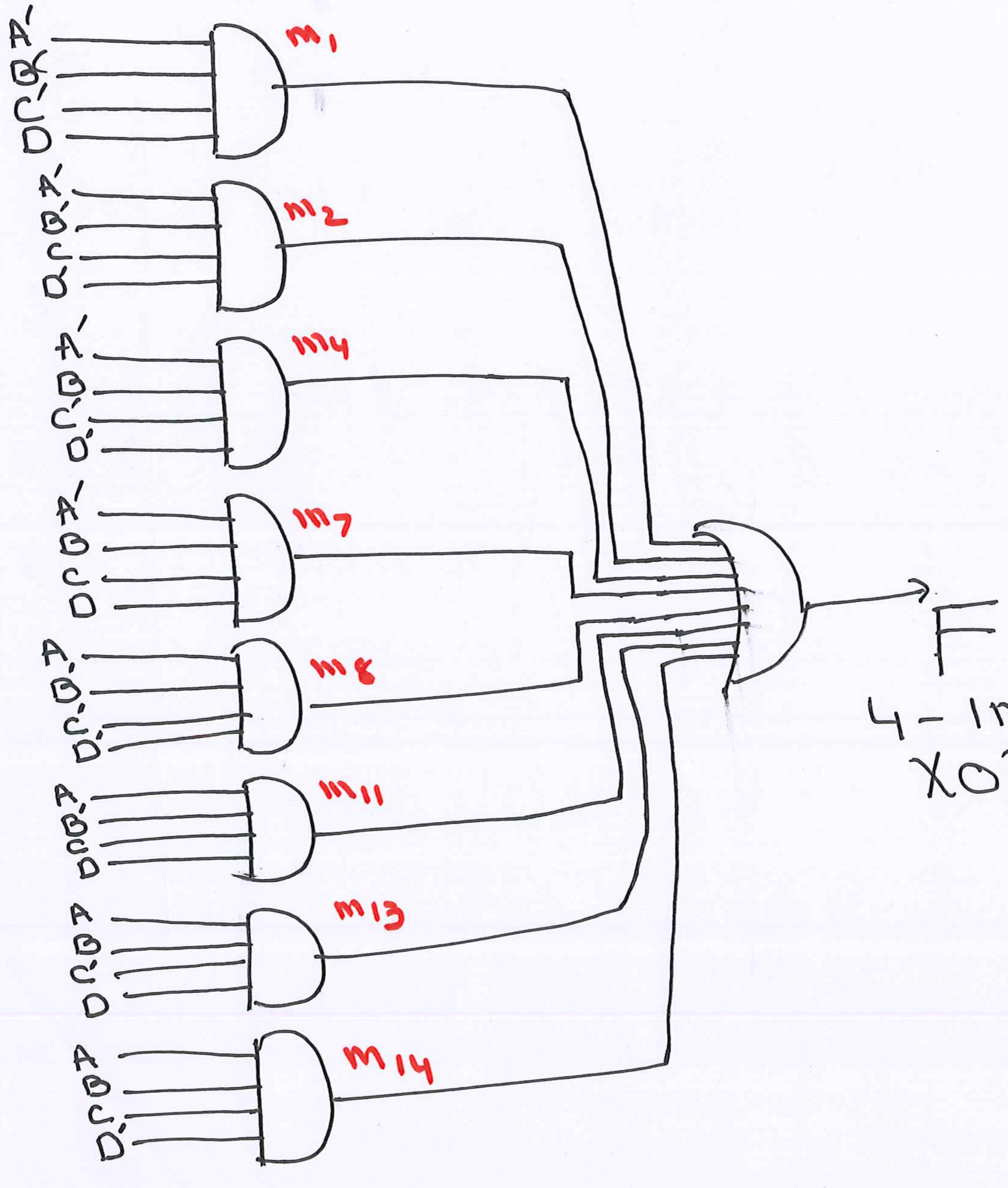
Question 3 [4 points]:

Implement a 4-input XOR function using AND, OR and NOT gates. Assume inputs as, A, B, C, D.

AB \ CD	00	01	11	10
00	m_0	m_1: 1	m_3	m_2: 1
01	m_4: 1	m_5	m_7: 1	m_6
11	m_{12}	m_{13}: 1	m_{15}	m_{14}: 1
10	m_8: 1	m_9	m_{11}: 1	m_{10}

4-input XOR

	A	B	C	D	F
m_0	0	0	0	0	0
✓m_1	0	0	0	1	1
✓m_2	0	0	1	0	1
m_3	0	0	1	1	0
✓m_4	0	1	0	0	1
m_5	0	1	0	1	0
m_6	0	1	1	0	0
✓m_7	0	1	1	1	1
✓m_8	1	0	0	0	1
m_9	1	0	0	1	0
m_{10}	1	0	1	0	0
✓m_{11}	1	0	1	1	1
m_{12}	1	1	0	0	0
✓m_{13}	1	1	0	1	1
✓m_{14}	1	1	1	0	1
m_{15}	1	1	1	1	0



4-Input XOR Function